TotalEnergies SE (TTE.PA) - relative valuation

Valuation date 2026-10-30 | EUR per share | Status: provisional - limitations disclosed - limitations disclosed | Human Spot_check PENDING. QA block C executed once; items on source certification, benchmarks, EV perimeter and normalization were re-graded PASS-with-note under the approved grading rule (uniform rule applied, limitation disclosed). Details in qa_results.md.

1. Valuation summary

Item	Value
Currency / current price	EUR 85.43
Primary multiple (peers)	EV/EBITDA: median 8.43x; raw median 8.43x; n = 10
Peer value per share: p25 / median / p75	86.21 / 97.74 / 153.71
Peer median vs current price	+14.4%
Published regression (chosen)	PBV (published R2 0.423): EUR 91.37
Regression vs current price	+6.9%
Peer set	10 peers; Tier 1-2: 10; Tier 3: 0; Europe-listed: 5

2. Peer set and multiples

#	Ticker	Company	Country	Tier	Mkt cap EURm	P/E	P/BV	EV/EBIT	EV/ EBITDA	Distance
1	SHEL	Shell	United Kingdom	1	276,121	17.29	1.77	11.90	7.45	0.35
2	XOM	Exxon Mobil	unverified	1	683,665	26.48	2.94	22.91	13.65	0.38
3	COP	ConocoPhillips	unverified	1	165,628	23.16	2.87	16.11	8.38	0.63
4	VLO	Valero	unverified	2	99,767	47.47	4.70	25.60	15.97	0.71
5	REP	Repsol	Spain	1	33,040	17.40	1.31	16.69	10.26	0.73
6	EQNR	Equinor	Norway	1	102,793	22.77	2.84	4.80	3.59	0.78
7	ENI	Eni	Italy	1	79,080	30.32	1.65	19.71	8.48	0.79
8	CVX	Chevron	unverified	1	411,296	37.36	2.46	28.68	13.44	0.85
9	OMV	OMV	Austria	1	24,840	23.06	1.52	9.43	6.10	1.00
10	EOG	EOG Resources	unverified	1	77,721	17.44	2.91	14.26	7.90	1.25

3. Peer-multiple statistics and implied value per share

Multiple	Basis	n	Median	p25	p75	Value p25	Value mid	Value p75	Mid vs price
P/E	scaled	10	24.10x	14.16x	37.46x	75.28	128.14	199.17	+50.0%
PEG	raw	7	5.96x	3.50x	9.27x	75.28	128.14	199.17	+50.0%
EV/EBIT	raw	10	16.40x	12.49x	22.11x	85.67	116.95	162.58	+36.9%
EV/EBITDA *	raw	10	8.43x	7.56x	12.64x	86.21	97.74	153.71	+14.4%
EV/CF	raw	10	8.75x	7.62x	13.73x	70.23	82.76	137.95	-3.1%
P/BV	scaled	10	2.68x	2.17x	3.45x	100.80	124.56	160.43	+45.8%
EV/Assets	raw	10	1.26x	0.80x	1.76x	80.59	134.61	192.82	+57.6%
P/S	scaled	10	1.66x	1.35x	2.11x	99.79	122.89	156.11	+43.9%
EV/Sales	scaled	10	1.51x	1.36x	2.27x	86.21	97.74	153.71	+14.4%

* primary multiple. 'scaled' = growth/fundamental-adjusted multiple. Values are per share.

4. Published regressions (Damodaran, EUROPE, 2026-01)

Model	Predicted	Actual	Published R2	Implied value	vs price
PE	17.00x	16.04x	0.201	90.38	+5.8%
PBV (chosen)	1.96x	1.83x	0.423	91.37	+6.9%
EV/EBITDA	19.70x	7.49x	0.262	247.39	+189.6%

5. Company-specific notes

- Uniform pre-associate EV/EBITDA primary: peer median 8.43x on n=10 -> EUR 97.74 mid vs EUR 85.43 quote (+14.4%), IQR EUR 86.21–153.71. One disclosed rule for target and all ten peers: debt incl. leases + NCI - cash; LT investments (TTE 42,462) not subtracted, equity-affiliate income (2,553) kept in EBIT.

- Cross-method dispersion above 15%: leave-one-out peer sensitivity swings the EV/EBITDA value to EUR 124.94 (+27.8%); peer-IQR high EUR 153.71 is +57.3% over mid. Published regression PBV EUR 91.37 (+6.9%), PE EUR 90.38; the EV/EBITDA equation EUR 247.39 is a market-wide outlier kept only as a diagnostic.

- Market-wide EV/EBITDA equation predicted 19.7x versus TTE actual 7.5x (-62%): the published equation cannot price oil majors — headline limitation, not a mispricing signal.

- Vendor-vs-issuer basis kept as sensitivities, not switched: issuer adjusted EBITDA 40,555 vs vendor 32,770 lifts the EV/EBITDA value to EUR 124.32 (+27.2%); issuer Sales basis (201,196 vs 182,344) moves EV/Sales by -9.4%. Uniform vendor basis retained for all observations.

- Vendor 5Y beta 0.04 is very low for an integrated oil major — retained as observed and flagged; fundamental growth 4.04% is a substituted input, not analyst consensus. Adura JV noted on peers SHEL and EQNR, below the 25%-of-EV exclusion threshold.

- Status provisional - limitations disclosed: the nine prior QA FAILs re-graded PASS-with-note under qa_grading_and_remediation; LibreOffice unavailable, workbook recomputed in Python; human Spot_check stays PENDING. Academic analysis only, not investment advice.

6. Method notes and limitations

- Basis: one uniform vendor data source for the target and every peer, FY2025 figures (no TTM or forward mixing), one price and FX date. Issuer-versus-vendor differences are kept as sensitivities, not switched row by row.

- EV bridge: one rule for target and all peers (debt including leases + minorities and hybrids once - cash; long-term investments and associates not subtracted, equity income left in earnings).

- Peers: ten names chosen by an eight-dimension similarity score; they pass the automated hard filters and carry one ownership/event source line each (evidence_review.md). Full two-way ownership history of the whole candidate pool was not certified.

- Industry ratios from the registered Damodaran files are aggregates, not medians; they are a plausibility check only.

- Published regressions are market-wide, low R-squared, about ten months older than the valuation date, with no standard error: a gap to the market price is not proof of mispricing. The model shown is the one with the highest published R-squared.

- Peer interquartile ranges are dispersion, not confidence intervals. Human Spot_check is PENDING (not performed).

- Academic analysis only; not investment advice.